

WMDR2-core

Package in package 'Model'

WMDR2-core

Version 0.1.10 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-13

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

wmdr2-core

Notes

1. temporalExtent shall be realized as time:interval

2. temporal<Name> elements other than temporalGeometry shall be realized as arrays of dictionary objects for (begin) date, and the values of <Name>, e.g.,

```
temporalTerritory = [{
```

```
"date": "2000-01-01",
```

```
"territory": "<ISO1>"
```

```
},
```

```
{
```

```
"date" 2024-07-13",
```

```
"territory": "<ISO2>"
```

```
},
```

3. Following the OGC Features model, geometry = geolocation = coordinates + elevation

4. Schedules

v0.1.4 Introduce instrument class.

v0.1.5 Expand instrument class.

v0.1.6 introduce more elements

v0.1.8

- require and expanded observation.observedDomain (includes domain, domainFeature [0..1], featureName [0..1])

- introduce temporalGeometry.methods

v0.1.9

- introduce deployment.temporalGeometry [0..1], verticalDistanceFromReferenceSurface [0..1]

- rename localReferenceSurface to referenceSurface

- rename observation.observedVariable to observation.observedProperty

- remove xType suffix to manage expectations: many elements are 'Types', i.e., codelist-based

v0.1.10

- rename environment.temporalTopographyBathymetry to topographyBathymetry

- add environment.temporalSurfaceRoughness

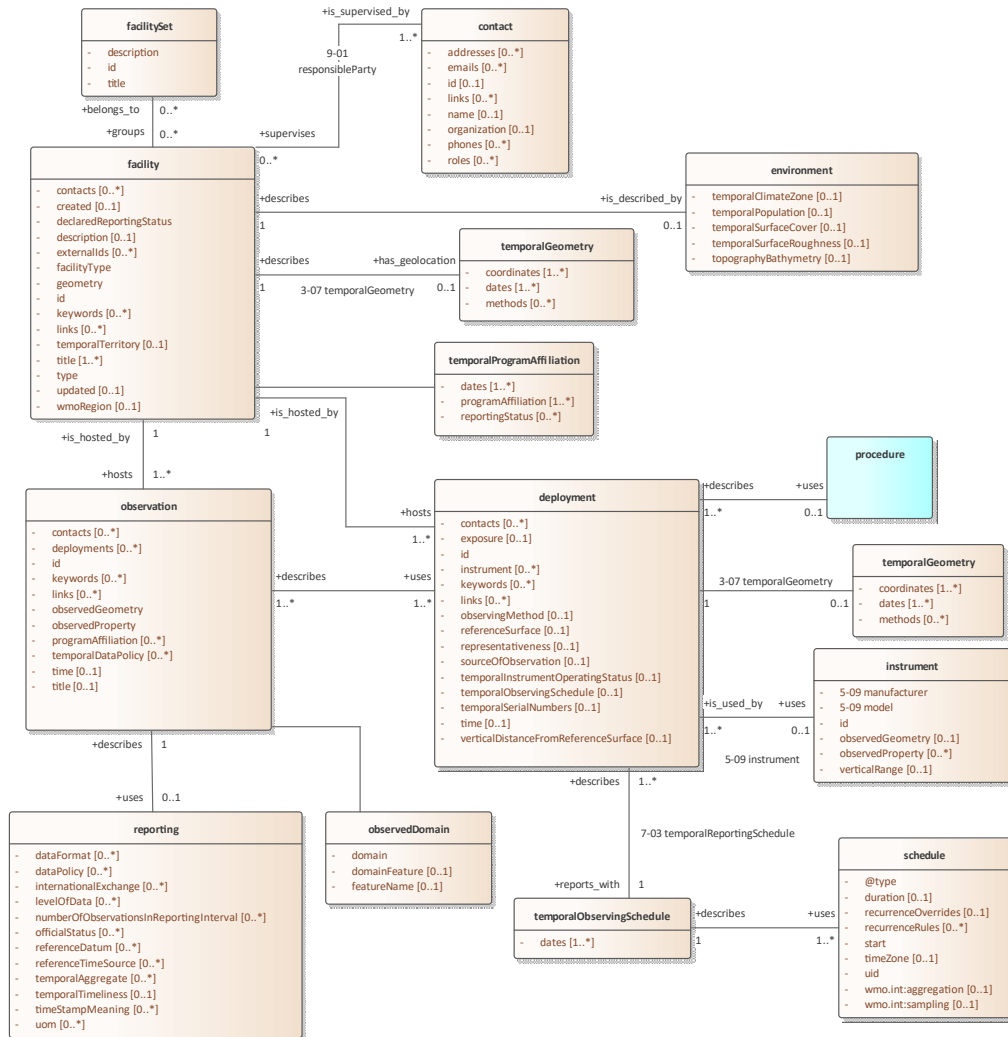
- rename instrumentCapability to instrument

- rename instrument.observableVariable to observedProperty, observableGeometry to observedGeometry.

WMDR2-core

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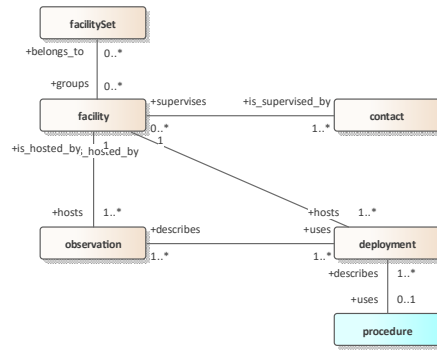


Figure 2: WMDR2-core-logical

aggregation

Class in package 'WMDR2-core'

WMDR-specific aggregation metadata stored under `schedule["wmo.int:aggregation"]`.

aggregation

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
🔹 diurnalBaseTime : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional base time for aggregate alignment, HH:MM:SS. [Is static True. Containment is Not Specified.]	
🔹 spatialResolution : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional spatial resolution; prefer numeric metres when known. [Is static True. Containment is Not Specified.]	
🔹 statistics : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Optional aggregate statistic values: mean, median, min, max, sum. [Is static True. Containment is Not Specified.]	
🔹 temporalAggregate : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Temporal aggregate period, e.g. PT1H, PID, P1M. [Is static True. Containment is Not Specified.]	

contact

Class in package 'WMDR2-core'

WMDR2 core JSON class: contact.

contact

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-02

ATTRIBUTES	
 addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Postal addresses.	[Is static True. Containment is Not Specified.]
 emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Email contacts.	[Is static True. Containment is Not Specified.]
 id : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional stable contact identifier.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Online resources associated with the contact.	[Is static True. Containment is Not Specified.]
 name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard)	[Is static True. Containment is Not Specified.]
 organization : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss	[Is static True. Containment is Not Specified.]
 phones : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Phone contacts.	[Is static True. Containment is Not Specified.]

ATTRIBUTES

◆ roles : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Specific contact roles such as owner, pointOfContact, principalInvestigator.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

✎ Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

deployment

Class in package 'WMDR2-core'

WMDR2 core JSON class: deployment.

deployment

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES

◆ contacts : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Deployment-level contacts.

[Is static True. Containment is Not Specified.]

◆ exposure : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Compact exposure classification value.

[Is static True. Containment is Not Specified.]

◆ id : Private

Deployment identifier; source XML id is preserved when available.

[Is static True. Containment is Not Specified.]

◆ instrument : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

References to reusable instrument ids.

[Is static True. Containment is Not Specified.]

◆ keywords : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
Deployment keywords.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Deployment links.	[Is static True. Containment is Not Specified.]
 observingMethod : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Realized as an array of dictionaries, i.e. key-value pairs: key: observingMethod id values: observation id	[Is static True. Containment is Not Specified.]
 referenceSurface : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Linked to heightAboveReferenceSurface	[Is static True. Containment is Not Specified.]
 representativeness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Compact representativeness value.	[Is static True. Containment is Not Specified.]
 sourceOfObservation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Compact source-of-observation code value.	[Is static True. Containment is Not Specified.]
 temporalInstrumentOperatingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Wrapper with instrumentOperatingStatus and dates arrays.	[Is static True. Containment is Not Specified.]
 temporalObservingSchedule : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) complex type: when are observations made? Should be expressed as a temporal coverage, e.g. "from Feb-Oct on working days from 06-18 UTC"	[Is static True. Containment is Not Specified.]
 temporalSerialNumbers : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Implemented as	

ATTRIBUTES

```
"temporalInstrumentSerialNumber": {
  "serialNumber": [...],
  "datetimes": [...],
}
```

[Is static True. Containment is Not Specified.]

 **time** : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Deployment temporal extent, date-resolution interval or null.

[Is static True. Containment is Not Specified.]

 **verticalDistanceFromReferenceSurface** : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified) 3-07 temporalGeometrySource: Public (Class) deployment
Cardinality: [1]Target: Public (Class) temporalGeometry
Cardinality: [0..1]

describes

has_geolocatio

 Association (direction: Unspecified) 7-03 temporalReportingScheduleSource: Public describes (Class) deployment
Cardinality: [1..*]Target: Public reports_with (Class)
temporalObservingSchedule
Cardinality: [1] Association (direction: Unspecified) 5-09 instrumentSource: Public is_used_by (Class) deployment
Cardinality: [1..*]Target: Public uses (Class) instrument
Cardinality: [0..1] Association (direction: Unspecified)Source: Public describes (Class) deployment
Cardinality: [1..*]Target: Public uses (Class) procedure
Cardinality: [0..1]

ObservingProcedure(s)

 Association (direction: Unspecified) 5-09 instrumentSource: Public describes (Class) deployment
Cardinality: [0..*]Target: Public uses (Class) instrument
Cardinality: [0..1] Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility

Target: Public hosts (Class) deployment

ASSOCIATIONS	
Cardinality: [1]	Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) observation Cardinality: [1..*]	Target: Public uses (Class) deployment Cardinality: [1..*]

environment

Class in package 'WMDR2-core'

Optional facility-environment object containing temporal climate, surface-cover, population, and topography/bathymetry histories.

environment

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-02. Last modified 2026-06-02

ATTRIBUTES	
 temporalClimateZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Temporal climate-zone history.	[Is static True. Containment is Not Specified.]
 temporalPopulation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) 	[Is static True. Containment is Not Specified.]
 temporalSurfaceCover : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Temporal surface-cover history.	[Is static True. Containment is Not Specified.]
 temporalSurfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) 	[Is static True. Containment is Not Specified.]
 topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) 	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) facility Cardinality: [1]	Target: Public is_described_by (Class) environment Cardinality: [0..1]

ASSOCIATIONS

facility

Class in package 'WMDR2-core'

Facility properties payload of the WMDR2 full record. Root Feature geometry/time are modeled on fullRecord, not here.

facility
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES

◆ contacts : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Facility-level contacts.

[Is static True. Containment is Not Specified.]

◆ created : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Record creation date/time metadata.

[Is static True. Containment is Not Specified.]

◆ declaredReportingStatus : Private

[Is static True. Containment is Not Specified.]

◆ description : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Facility description.

[Is static True. Containment is Not Specified.]

◆ externalIds : Private
Alias: 3-06 (primary & additional) WSIs
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

External identifiers such as WMO:WIGOS.

[Is static True. Containment is Not Specified.]

◆ facilityType : Private
Alias: 3-04 facilityType

Compact WMDR facility type code value.

[Is static True. Containment is Not Specified.]

◆ geometry : Private
Alias: 3-07 current geolocation

geometry = geolocation = coordinates + elevation

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
💠 id : Private Alias: 3-06 (primary) WSI to prefixed with name space "facility:"	[Is static True. Containment is Not Specified.]
💠 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON.	[Is static True. Containment is Not Specified.]
💠 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links.	[Is static True. Containment is Not Specified.]
💠 temporalTerritory : Private Alias: 3-02 temporal Territory Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities	[Is static True. Containment is Not Specified.]
💠 title : Private Alias: 3-03 facilityName Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) properties.title = 3-03 facilityName	[Is static True. Containment is Not Specified.]
💠 type : Private Facility properties type; const facility.	[Is static True. Containment is Not Specified.]
💠 updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata.	[Is static True. Containment is Not Specified.]
💠 wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public describes (Class) facility Cardinality: [1]	Target: Public has_geolocation (Class) temporalGeometry Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) deployment Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public groups (Class) facility Cardinality: [0..*]	Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]
 Association (direction: Unspecified) 9-01 responsibleParty	
Source: Public supervises (Class) facility Cardinality: [0..*]	Target: Public is_supervised_by (Class) contact Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public (Class) facility	Target: Public (Class) temporalProgramAffiliation
 Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observation Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) facility Cardinality: [1]	Target: Public is_described_by (Class) environment Cardinality: [0..1]

facilitySet

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

facilitySet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES
 description : Private

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
 id : Private	[Is static True. Containment is Not Specified.]
 title : Private	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public groups (Class) facility Cardinality: [0..*]	Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]

fullRecord

Class in package 'WMDR2-core'

Root WMDR2 full-record Feature. This class represents the GeoJSON/OGC Records envelope around the facility-centric WMDR2 payload.

fullRecord

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 conformsTo : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Required conformance URIs including OGC Records core and WMDR2 full-record.	[Is static True. Containment is Not Specified.]
 geometry : Private Current GeoJSON geometry. Derived from the most recent temporalGeometry coordinate when available; may be null in JSON.	[Is static True. Containment is Not Specified.]
 id : Private Facility full-record identifier, e.g. facility:<WIGOS identifier>.	[Is static True. Containment is Not Specified.]
 properties : Private Facility properties payload.	[Is static True. Containment is Not Specified.]

ATTRIBUTES

temporalGeometry : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

WMDR2 facility coordinate history, kept at the root Feature level.

[Is static True. Containment is Not Specified.]

time : Private

OGC Records temporal extent of the facility resource. Date-resolution interval or null.

[Is static True. Containment is Not Specified.]

type : Private

GeoJSON Feature type; const Feature.

[Is static True. Containment is Not Specified.]

instrument

Class in package 'WMDR2-core'

WMDR2 core JSON class: instrument.

instrument

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-02

ATTRIBUTES

id : Private

Reusable instrument id.

[Is static True. Containment is Not Specified.]

manufacturer : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

This should come from a registry/catalogue, but should accept any organization.

[Is static True. Containment is Not Specified.]

model : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

This should come from a registry/catalogue, but can accept "custom-built".

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment

Target: Public uses (Class) instrument

ASSOCIATIONS

Cardinality: [0..*]

Cardinality: [0..1]

instrument

Class in package 'WMDR2-core'

Describes instrument capabilities, not a specific instrument instance.

For semantic consistency, terms like `observedProperty` and `observedGeometry` are used, even though they just describe the instrument capabilities, not the actual observations. With this, instrument becomes re-usable between deployments and even facilities.

instrument

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-05-22. Last modified 2026-06-13

ATTRIBUTES

5-09 manufacturer : Private

[Is static True. Containment is Not Specified.]

5-09 model : Private

[Is static True. Containment is Not Specified.]

id : Private

[Is static True. Containment is Not Specified.]

observedGeometry : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Specify using <http://codes.wmo.int/wmdr/Geometry>

[Is static True. Containment is Not Specified.]

observedProperty : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Specify as an array of values taken from <http://codes.wmo.int/wmdr/ObservedVariable> or as free text description.

[Is static True. Containment is Not Specified.]

verticalRange : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 5-09 instrument

Source: `Public is_used_by` (Class) deploymentTarget: `Public uses` (Class) instrument

ASSOCIATIONS

Cardinality: [1..*]

Cardinality: [0..1]

observation

Class in package 'WMDR2-core'

WMDR2 core JSON class: observation.

observation

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES



contacts : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Observation-level contacts.

[Is static True. Containment is Not Specified.]



deployments : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Deployment id references.

[Is static True. Containment is Not Specified.]



id : Private

Observation identifier, e.g. observation:<observedVariable>.

[Is static True. Containment is Not Specified.]



keywords : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Observation keywords.

[Is static True. Containment is Not Specified.]



links : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Observation links.

[Is static True. Containment is Not Specified.]



observedGeometry : Private

cf. <http://codes.wmo.int/wmdr/Geometry>

[Is static True. Containment is Not Specified.]



observedProperty : Private

Alias: OMS:ObservedProperty, iop:Variable

ATTRIBUTES

The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2.

NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific.

Examples include:

- observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...)
- objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll
- observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.
- matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ...

These elements will be supported by code lists from codes.wmo.int/wmdr2.

Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc.

[Is static True. Containment is Not Specified.]

 programAffiliation : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Program-affiliation timeline fragments inherited from WMDR1 observations.

[Is static True. Containment is Not Specified.]

 temporalDataPolicy : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Optional temporal data-policy history.

[Is static True. Containment is Not Specified.]

 time : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Observation temporal extent, date-resolution interval or null.

[Is static True. Containment is Not Specified.]

 title : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Human-readable title derived from domain, geometry, observed variable, and optional label.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) observation

Target: Public (Class) observedDomain

 Association (direction: Unspecified)

ASSOCIATIONS	
Source: Public describes (Class) observation Cardinality: [1..*]	Target: Public uses (Class) deployment Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) observation Cardinality: [1]	Target: Public uses (Class) reporting Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observation Cardinality: [1..*]

observedDomain

Class in package 'WMDR2-core'

observedDomain

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES
 domain : Private cf. http://codes.wmo.int/wmdr/Domain [Is static True. Containment is Not Specified.]
 domainFeature : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Concept motivated by hydrology, but may be useful elsewhere. Propose new code list, with terms like - river basin - lake - river - catchment - canopy - boundary layer - thermocline - ... [Is static True. Containment is Not Specified.]
 featureName : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) free text field, e.g., "Danube Basin" [Is static True. Containment is Not Specified.]

ASSOCIATIONS

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) observation

Target: Public (Class) observedDomain

procedure

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

procedure

Version 1.0 Phase 1.0 Proposed

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ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) deployment

Cardinality: [1..*]

Target: Public uses (Class) procedure

Cardinality: [0..1]

ObservingProcedure(s)

recurrenceRule

Class in package 'WMDR2-core'

WMDR2 core JSON class: recurrenceRule.

recurrenceRule

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES

 @type : Private

JSCalendar RecurrenceRule type; const RecurrenceRule.

[Is static True. Containment is Not Specified.]

 byDay : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Optional day constraints.

[Is static True. Containment is Not Specified.]

 byHour : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Optional hour constraints.

[Is static True. Containment is Not Specified.]

ATTRIBUTES	
byMinute : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Optional minute constraints.	[Is static True. Containment is Not Specified.]
byMonth : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Optional month constraints.	[Is static True. Containment is Not Specified.]
count : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional recurrence count.	[Is static True. Containment is Not Specified.]
frequency : Private JSCalendar recurrence frequency.	[Is static True. Containment is Not Specified.]
interval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional recurrence interval.	[Is static True. Containment is Not Specified.]
until : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional recurrence end.	[Is static True. Containment is Not Specified.]

reporting

Class in package 'WMDR2-core'

Observation reporting metadata represented as aligned arrays. temporalTimeliness is itself a temporal wrapper.

reporting
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES	
Aligned array of compact data-format values.	[Is static True. Containment is Not Specified.]
 dataPolicy : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of data-policy objects, preserving attribution when present.	[Is static True. Containment is Not Specified.]
 internationalExchange : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of international-exchange flags.	[Is static True. Containment is Not Specified.]
 levelOfData : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of compact level-of-data values.	[Is static True. Containment is Not Specified.]
 numberOfObservationsInReportingInterval : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array copied from WMDR reporting metadata when present.	[Is static True. Containment is Not Specified.]
 officialStatus : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of official-status flags.	[Is static True. Containment is Not Specified.]
 referenceDatum : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of reference-datum objects.	[Is static True. Containment is Not Specified.]
 referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of reference-time-source values.	[Is static True. Containment is Not Specified.]
 temporalAggregate : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of reporting/aggregation periods, renamed from temporalReportingInterval.	[Is static True. Containment is Not Specified.]

ATTRIBUTES

 temporalTimeliness : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Optional wrapper with timeliness durations and dates.

[Is static True. Containment is Not Specified.]

 timeStampMeaning : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Aligned array of timestamp-meaning values.

[Is static True. Containment is Not Specified.]

 uom : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Aligned array of compact unit code values.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1]

Target: Public uses (Class) reporting
Cardinality: [0..1]

schedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

schedule

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES

 @type : Private

JSCalendar object type; const Event.

[Is static True. Containment is Not Specified.]

 duration : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Occurrence duration describing coverage window or sampling duration.

[Is static True. Containment is Not Specified.]

 recurrenceOverrides : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
JSCalendar recurrence override patches; null override values are invalid.	[Is static True. Containment is Not Specified.]
 recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence rules.	[Is static True. Containment is Not Specified.]
 start : Private Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.	[Is static True. Containment is Not Specified.]
 timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) IANA time zone or UTC default.	[Is static True. Containment is Not Specified.]
 uid : Private JSCalendar-safe schedule uid.	[Is static True. Containment is Not Specified.]
 wmo.int:aggregation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDR-specific aggregation metadata.	[Is static True. Containment is Not Specified.]
 wmo.int:sampling : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDR-specific sampling metadata preserved from dataGeneration.sampling.	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public describes (Class) temporalObservingSchedule Cardinality: [1]	Target: Public uses (Class) schedule Cardinality: [1..*]

temporalClimateZone

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalClimateZone.

temporalClimateZone
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
🔹 climateZone : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Compact climate-zone values. [Is static True. Containment is Not Specified.]	
🔹 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with climateZone. [Is static True. Containment is Not Specified.]	

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-02

ATTRIBUTES	
🔹 coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
🔹 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates. [Is static True. Containment is Not Specified.]	
🔹 methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
✍ Association (direction: Unspecified) 3-07 temporalGeometry Source: Public describes (Class) facility Target: Public has _geolocation (Class)	

ASSOCIATIONS

Cardinality: [1]

temporalGeometry

Cardinality: [0..1]

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-12. Last modified 2026-06-02

ATTRIBUTES



coordinates : Private

Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Coordinate history array; each entry is lon, lat[, elevation].

[Is static True. Containment is Not Specified.]



dates : Private

Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Date array aligned with coordinates.

[Is static True. Containment is Not Specified.]



methods : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

use <http://codes.wmo.int/wmdr/GeopositioningMethod>

[Is static True. Containment is Not Specified.]

ASSOCIATIONS



Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public (Class) deployment

Cardinality: [1]

Target: Public (Class) temporalGeometry

Cardinality: [0..1]

describes

has_geolocatio

temporalInstrumentOperatingStatus

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalInstrumentOperatingStatus.

temporalInstrumentOperatingStatus

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with instrumentOperatingStatus. [Is static True. Containment is Not Specified.]	
 instrumentOperatingStatus : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Compact instrument-operating-status values. [Is static True. Containment is Not Specified.]	

temporalObservingSchedule

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalObservingSchedule.

temporalObservingSchedule
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-02

ATTRIBUTES	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with observingSchedule. [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) temporalObservingSchedule Cardinality: [1]	Target: Public uses (Class) schedule Cardinality: [1..*]
 Association (direction: Unspecified) 7-03 temporalReportingSchedule	
Source: Public describes (Class) deployment Cardinality: [1..*]	Target: Public reports_with (Class) temporalObservingSchedule Cardinality: [1]

temporalPopulation

Class in package 'WMDR2-core'

temporalPopulation
Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-02. Last modified 2026-06-02

ATTRIBUTES	
 dates : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Begin dates aligned with the corresponding values. [Is static True. Containment is Not Specified.]	
 populationDensity : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Aligned population-density values. The corresponding begin dates are in dates[]. [Is static True. Containment is Not Specified.]	

temporalProgramAffiliation

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalProgramAffiliation.

temporalProgramAffiliation

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with programAffiliation. [Is static True. Containment is Not Specified.]	
 programAffiliation : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Compact program-affiliation values. [Is static True. Containment is Not Specified.]	
 reportingStatus : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Optional compact reporting-status values aligned with programAffiliation. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) facility Target: Public (Class) temporalProgramAffiliation	

temporalSerialNumbers

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalSerialNumbers.

temporalSerialNumbers
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with serialNumber. [Is static True. Containment is Not Specified.]	
 serialNumber : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Serial-number history values. [Is static True. Containment is Not Specified.]	

temporalSurfaceCover

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalSurfaceCover.

temporalSurfaceCover
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with surfaceCover. [Is static True. Containment is Not Specified.]	
 surfaceCover : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Compact surface-cover values. [Is static True. Containment is Not Specified.]	

temporalTerritory

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalTerritory.

temporalTerritory
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with territory.	[Is static True. Containment is Not Specified.]
territory : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Compact territory values.	[Is static True. Containment is Not Specified.]

temporalTimeliness

Class in package 'WMDR2-core'

Temporal wrapper for reporting latency/delay durations.

temporalTimeliness
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with timeliness.	[Is static True. Containment is Not Specified.]
timeliness : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Reporting latency/delay durations.	[Is static True. Containment is Not Specified.]

temporalTopographyBathymetry

Class in package 'WMDR2-core'

temporalTopographyBathymetry
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-02. Last modified 2026-06-02

ATTRIBUTES

ATTRIBUTES

◆ dates : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Begin dates aligned with the corresponding values.

[Is static True. Containment is Not Specified.]

◆ topographyBathymetry : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Topography/bathymetry values using the elements available in the WMDR source XML.

[Is static True. Containment is Not Specified.]